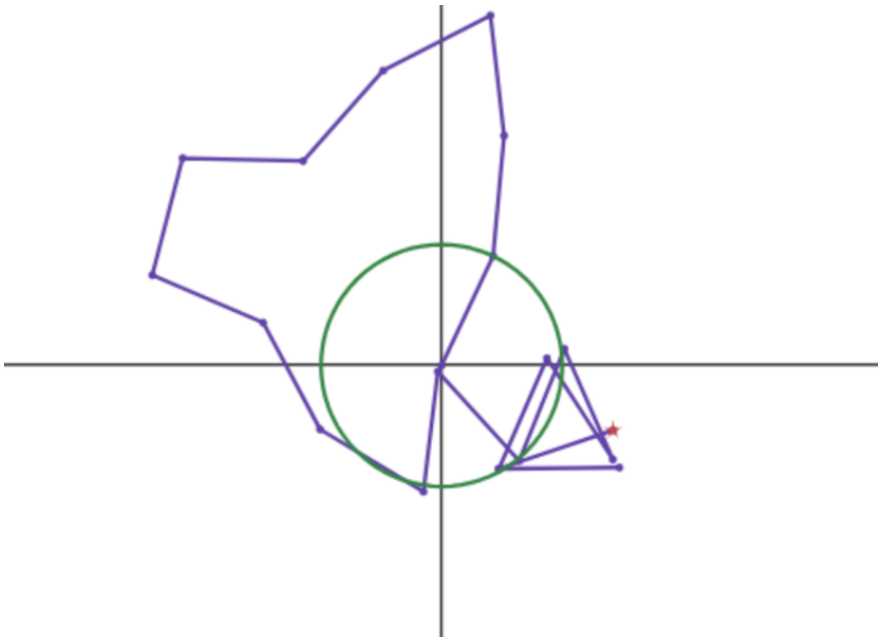
IB Mathematics Analysis and Approaches Higher Level

Internal Assessment



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1 Introduction

It was difficult to settle on a topic for this project, because there is so much richness in math. Firstly, how does one settle on just one topic of interest when almost anything will satisfy that constraint? Math has, for years now, been my favorite subject in and out of school. And then, how can that topic then be formulated such that it will prove to be appropriate for the constraints an Internal Assessment creates; that is, where is the balance between too simple and too complicated? I recognize in myself a tendency to over complicate, ideally in pursuit

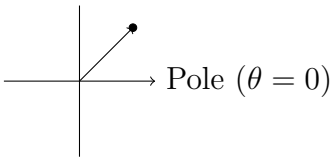
of making amazing discoveries or connections, in projects like this. The self-directedness and open-endedness of the investigation (which is perhaps the same open-endedness inherent to much of mathematical research and inquiry - but this is my closest experience with it) lends to difficulty in needing to choose something, but also can inspire specific questions of considerable interest, or at least I have found that to be the case here. The question and topic of this investigation I formed out of that struggle of trying to find something that interests me, fits the scope of the Internal Assessment, and with an attainable aim.

1.1 Polar and Euler Form of Complex Numbers[1]

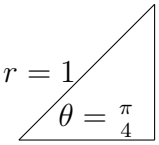
Before I state the aim of this paper, however, I wish to explain and define polar and Euler forms of complex numbers, as they will be a key element of what is to come. These, and following, definitions are necessary to the investigation for two main reasons: (1) it follows in the mathematical and philosophical precedent of establishing rigor by first ensuring the object of your study is properly and explicitly defined, so as to avoid pitfalls and fallacies that were the cause of simply not knowing what you are talking about, and (2) this is a problem that originated with me; that is, I know of no formal or popular way to express my question, and so in order to properly articulate it, I must be clear.

I will take complex numbers of the form $z = a + bi,\ a, b \in \mathbb{R}$, and the modulus and argument of complex numbers to be presupposed premises. For the purpose of this investigation, I want to be able to define complex numbers using their arguments, though, not the real-valued numbers a and b. If we conceive of a complex number, without yet formalizing it, that is so defined, the number system of complex numbers becomes an analog of polar coordinates. In the polar system, points are wholly defined by their polar radius r and their angle from the pole θ. Take the point $(r, \theta) = (1, \frac{\pi}{4})$, for example. Plotting this on the polar axes, I can produce this graph:

[1]This subsection paraphrases Haese et al. (2019)



Notice how the line drawn from the origin to the point can be used to make a right-angled triangle, as below:



Thus, the horizontal and vertical legs of the triangle are given as $(1)\cos(\frac{\pi}{4}) = (1)\sin(\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, respectively. If r is the distance from the origin to the point, then the lengths of the horizontal and vertical legs are the x and y coordinates of the point in a Cartesian form. Any polar number will have the same relationships to right triangles, and so, the Cartesian form of a polar point is $(x, y) = (r\cos(\theta), r\sin(\theta))$. As stated above, complex numbers are analogs of polar numbers, and so it holds true for complex numbers that the Cartesian form is $(x, y) = (|z|\cos(\theta), |z|\sin(\theta))$.

Now we take all this information back to the complex plane. Given the Cartesian form of a complex number, and the fact that a Cartesian x-coordinate relates to the real part and a Cartesian y-coordinate relates to the imaginary part of a complex number, we can write z as $|z|(\cos(\theta) + \sin(\theta)i)$. This is the basis of the polar form of a complex number, typically written as $z = |z|\,\mathrm{cis}(\theta)$, where $\mathrm{cis}(\theta) = \cos(\theta) + i\sin(\theta)$. Now, it is a product of Euler's genius that we know that $e^{i\theta} = \mathrm{cis}(\theta)$. Euler's Equation thus allows complex numbers to also be written as

$$z = |z|e^{i\theta},$$

which is the Euler form of complex numbers. I prefer the compactness of Euler form and will be using it for the detailing of the question of this investigation, but, as we will see, polar form will have its uses.

1.2 The Question

Consider a complex number z_1, defined in the Euler form $z_1 = e^{i\alpha}$. Suppose that α is a continuous random variable that uniformly takes any value in the interval $[0, 2pi)$, which allows every point on the unit circle $|z| = 1$ to be equally likely as a value of z_1; that is, z_1 is defined as a random point on the unit circle in the complex plane. To z_1 add another complex number that is defined as $z = e^{i\beta}$, where β is another continuous random variable that also uniformly takes any value in the domain $[0, 2pi)$, and suppose α and β are independent of each other. Let $z_2 = z_1 + e^{i\beta}$. I begin with the question, *what is the probability that z_2 falls within the unit circle in the complex plane $|z| = 1$?*

Then, I expand the investigation to finding the probability that a complex number made from adding n complex numbers defined similarly as above lies within the unit circle. Describing these numbers can become cumbersome for larger values of n, so to give a simplified, generalized, and exact definition of the complex numbers I will be dealing with, I see only the possibilities of turning to a recursive definition, where $z_n = z_{n-1} + e^{i\alpha_n}$, or something of that nature, or using summation. To include both, I define z_n as follows:

$$z_n = \sum_{x=1}^{n} e^{i\alpha_x} = z_{n-1} + e^{i\alpha_n},$$

where α_x is a continuous random variable that takes on a value in the interval $[0, 2\pi)$ for all x, and does so independently of all other values of α_x, and is the argument the nth complex number in the series. Notice that $z_n = \sum_{x=1}^{n} \text{cis}(\alpha_x)$, as well, using the aforementioned equivalent notations for complex numbers. Now, I give the ultimate question to be explored:

What is the probability that z_n falls within the unit circle $|z| = 1$ for a given n?

2 Investigation

I will perform this investigation by explaining models and simulations that I made for specific and general cases of the question and then moving on to finding exact solutions.

2.1 Modeling and Simulation

I began to explore this question by building means to visualize it with the Desmos graphing calculator. Desmos has the capability to plot complex numbers on an Argand plane, and, via the `random()` function, pseudo-random numbers in the interval $[0, 1)$ can be produced (Desmos, n.d.-b). The `random()` function will also produce a new pseudo-random number every time it is called, which means that Desmos can be used to plot an approximation of z_n. The randomness and independence of each α_x is approximated by `random()*2`π, but the model can help with visualization and building intuition for solving the problem.
So, after enabling `Complex Mode` in Desmos, I define z_n as $f(n)$ in the command:

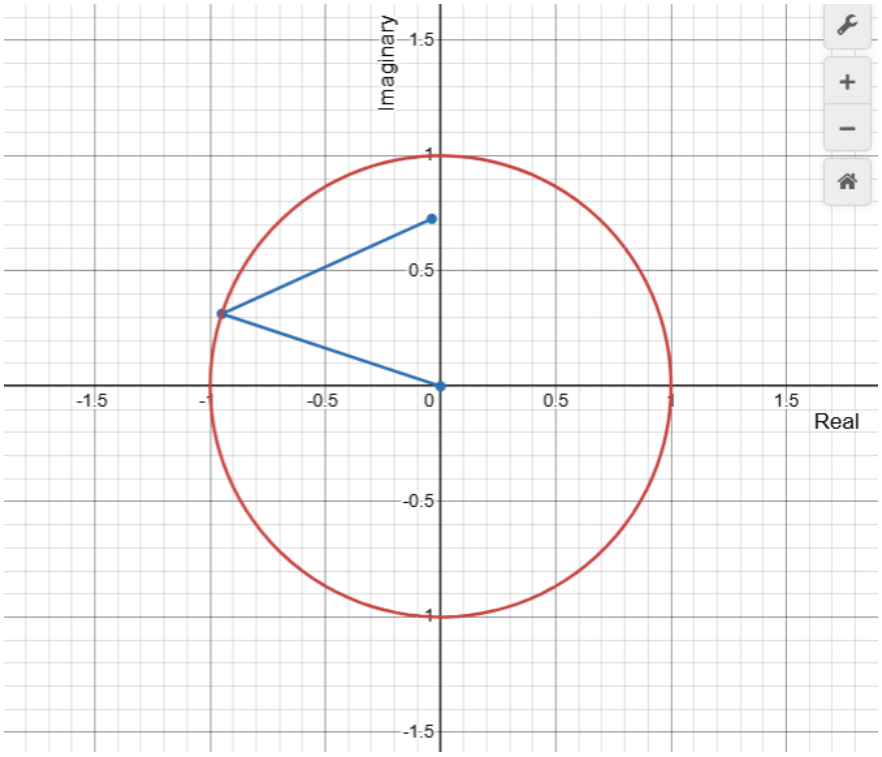
$$f(n) = e^{i \cdot 2\pi \cdot \text{random}(\ )} + f(n-1)$$

I opted to use the recursive definition because that will enable $f(n)$ to be expressed in a table, which allows a sequence a points to be connected by lines. I prefer the look of the graph with lines because it helps keep track of the order of points. Additionally, since $f(n)$ is defined recursively, there must be a base case. Since $z_1 = e^{i\alpha_1}$ (see §1.2), $f(1) = e^{i*\texttt{random()*2}\pi}$, $f(0)$ must be equal to zero, giving us the base case $f(0) = 0$. Next, a table must be constructed that saves values of $f(n)$ and allows the user to control which values of n are graphed. An example of the table where $n = 2$ is shown below:

n	$f(n)$
0	0
1	$-0.94933 + 0.31428i$
2	$-0.038093 + 0.72616i$

Notice how, even though $f(n)$ is defined in Euler form, Desmos automatically calculates the complex numbers in Cartesian form.

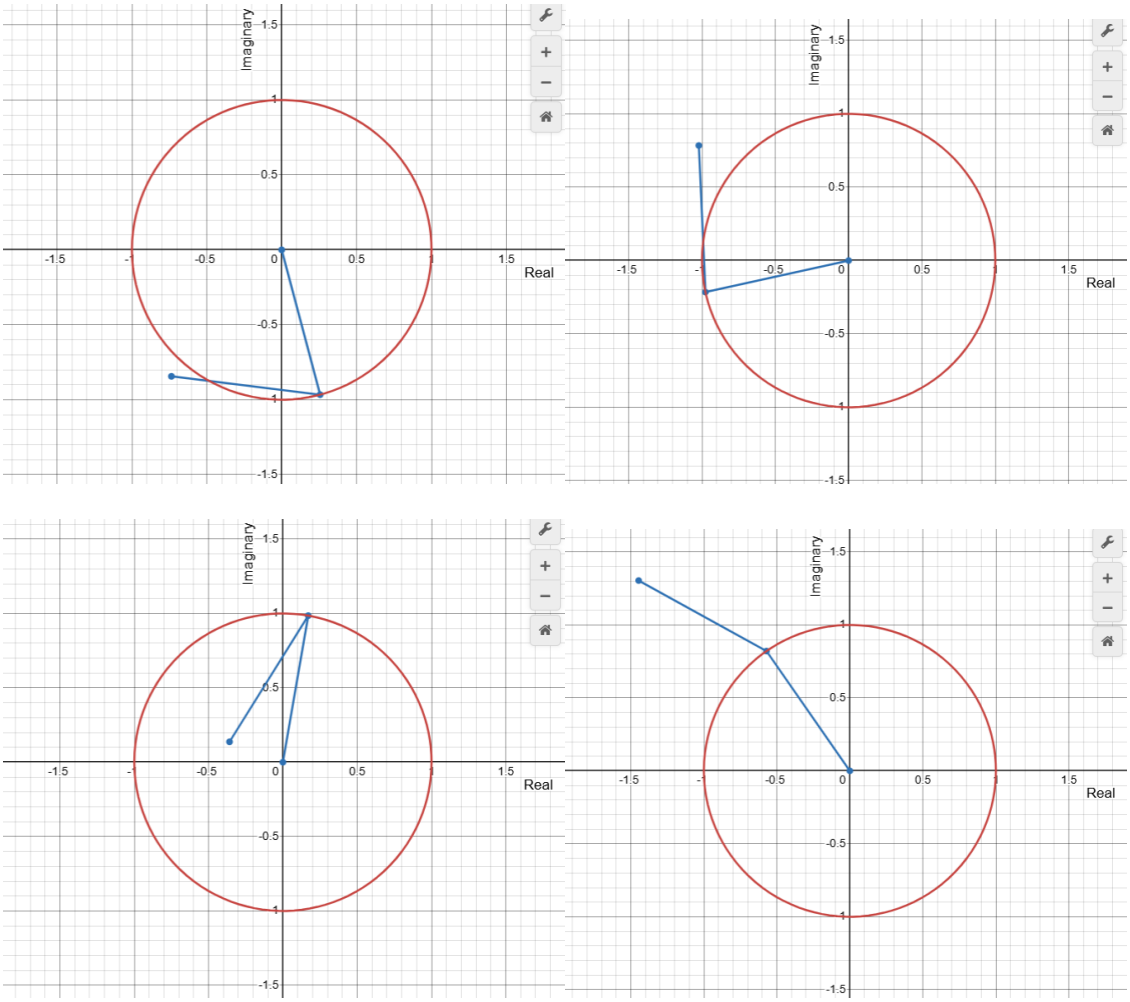
The last three steps in completing the graph are to enter the Desmos command `z=f(n)`, which actually plots the points, activate the lines for this new command in the Desmos settings, and graph the unit circle. Desmos cannot actually plot inputs like `|z|=1`, so we make do with the standard form $x^2 + y^2 = 1$. For some random output and $n = 2$, this yields the graph below.



I find this method of graphing what z_n actually is to be incredibly helpful in understanding what the question is asking. And actually, it was in part by making and experimenting with graphics and simulations like this that I thought of the question. The outputs shown in the above graph would be what I will begin to refer to as a "success", that is, z_n falls within the unit circle. At this stage it became clear to me that the real component of z_n is the sum of the cosines of the random angles and the imaginary component is the sines of the same angles. Then, by using the modulus or distance formula, the question could be reworded as follows if the definition of z_n is maintained:

What is the probability that $|z_n| < 1$, *where* $|z_n| = \sqrt{(\sum_{x=1}^{n} \cos(\alpha_x))^2 + (\sum_{x=1}^{n} \sin(\alpha_x))^2}$*?*

This more analytical phrasing would not have been helpful for initially building intuition, but will prove to be valuable for setting up simulations. At this moment, however, let us now use Desmos' ability to re-randomize the values of `random()` and make note of a few more possible outcomes for z_2. These are shown below.



Now, only one of these four graphs shows a success. A sample size of 5 is much too small to make any inference, but we can start to think that the probability of success for $n = 2$ is less than a half, which should make sense, because all of the possible values of z_2 will lie within a circle with radius 2 centered at the origin, and a success is when z_2 lands within the unit circle centered at the origin. The range of z_2 is so because when you add two complex numbers each with a modulus of 1, starting at the origin, the greatest possible modulus for the net complex number is 2, which would occur when the two complex numbers that make up z_2 have the same argument ($\alpha_1 = \alpha_2$), essentially making z_2 a radius of the circle $r = 2$. The range of z_2 is then the area within a circle with radius 2, so it would make sense that if each point within the range of z_2 was equally likely, the probability that $|z_2| < 1$ would be the area of a circle with radius 1 divided by the area of a circle with radius 2. The area of a

circle is $A = \pi r^2$, so this hypothetical (the hypothesis being that every possible value of z_2 is equally likely) probability would be $\dfrac{\pi(1)^2}{\pi(2)^2} = \frac{1}{4}$.

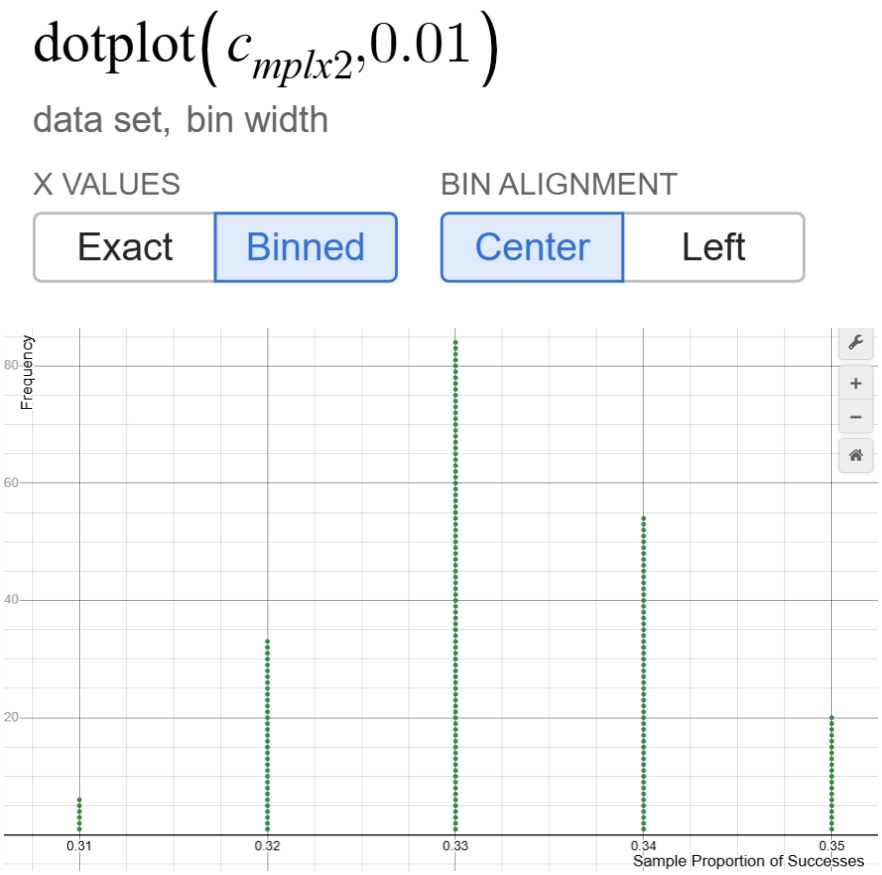
But the Desmos simulation does not actually count successes, so it is not an efficient way to estimate the probability of success. So, in continuing the process of making simulations until I have sufficient intuition, I wrote a series of Python programs that could be used to run relevant simulations more effectively, the first of which is to estimate this value for $n = 2$. In fact, let us only consider z_2 for some time, see what results that will yield, and then bring our attention to cases with higher n. Steven Strogatz writes that "hard problems become easier when they're split into chunks", and so $n = 2$ will be our first so-called chunk, with the intention of then eventually considering the whole problem (Strogatz, 2019, p. xv). And this is how I approached this problem initially, focusing on z_2 in hopes that results could be generalized or used as stepping-stones towards the answer.

For the exact code of this initial program (titled `IA1.py`), see §A.1 in the appendix, but to summarize how it works, it calculates z_2 1000 times and counts how many times $|z_2| < 1$, that is, how many times z_2 falls within the unit circle. And then, it divides this by 1000 to get the proportion of successes for this particular sample. The process is then repeated for a total of 50 samples, and all of the proportions of success for each are output. An example output is shown below:

```
PS C:\Users\samre\OneDrive\Desktop\project> python .\IA1.py
[0.3385, 0.3315, 0.34, 0.33, 0.32, 0.3285, 0.3365, 0.3455, 0.3345, 0.3265, 0.3355, 0.33
45, 0.341, 0.319, 0.3335, 0.3125, 0.3295, 0.3095, 0.35, 0.335, 0.339, 0.325, 0.322, 0.3
195, 0.347, 0.3355, 0.3225, 0.3355, 0.3225, 0.3485, 0.3285, 0.328, 0.338, 0.338, 0.3365
, 0.3345, 0.3185, 0.342, 0.334, 0.34, 0.3285, 0.331, 0.35, 0.3425, 0.319, 0.3195, 0.319
5, 0.3405, 0.3225, 0.3245]
```

We can plot these sample proportions as a dot plot and use it to estimate the true proportion - and thus probability - of success, but before we do so, notice that the program's output is clearly suggesting a value of $\frac{1}{3}$. I find that Desmos' statistical functions are very easy to work with, so we go back to Desmos to process the data. The relevant functions will be `dotplot()`, `mean()`, `median()`, `stdev()`, which will create a dot plot from the computer output, and find the mean and median of the data. The program IA1.py (§A.1) gives 50

sample proportions, but I will increase this to 200 (by changing the indexing variable `m` in the program) in order to increase the sample size of sample proportions in order make the mean and median based on the computer output better estimations of the true proportion of successes by reducing sampling variability. The function `dotplot()` requires a data set in the form of a list and a value that sets the bin width that Desmos will use in automatically binning the data (Desmos, n.d.-a). Most of the proportions the script gives go out to three decimal places (because they are proportions calculated by dividing by 1000), so the bin width will be set to `0.01`, binning the data based on the hundredth-place value. I then input the following command into Desmos, which produces the subsequent dot plot:



where c_{mplx2} is the list of data, named so because it is, again, the list of proportions of successes when the complex number is the sum of two complex numbers $e^{i\alpha_1}$ and $e^{i\alpha_2}$; that is, this is proportion of successes for a sample of values of z_2.

Desmos also gives the following results for the mean and median of the data set:

$$\text{mean}\left(c_{mplx2}\right) = 0.3327725$$

$$\text{median}\left(c_{mplx2}\right) = 0.332$$

There is now an abundance of evidence for the value of the proportion of successes when $n = 2$ being $\frac{1}{3}$, but how can that be verified? And, in mathematical exploration, we want to understand not only that something *is* true, but also *why* it is true, so why would $P(|z_2| < 1) = \frac{1}{3}$, if that hypothesis can indeed be verified?

2.2 Geometric Approach to finding $P(|z_2| < 1)$

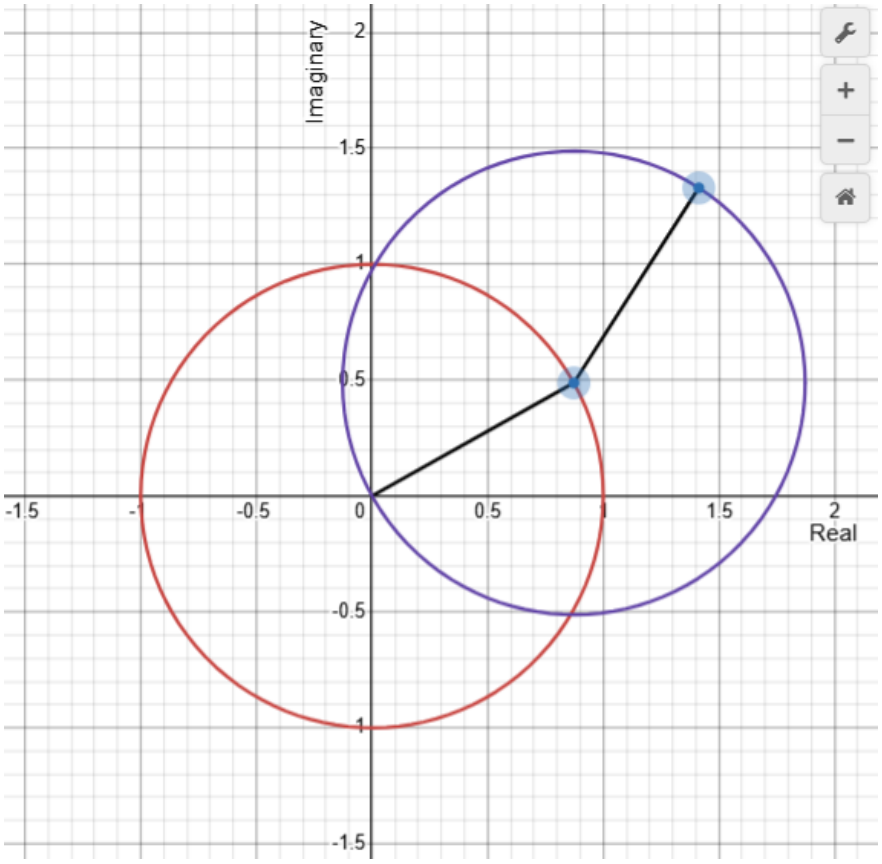
Recall once more the definition of z_n:

$$z_n = \sum_{x=1}^{n} e^{i\alpha_x}$$

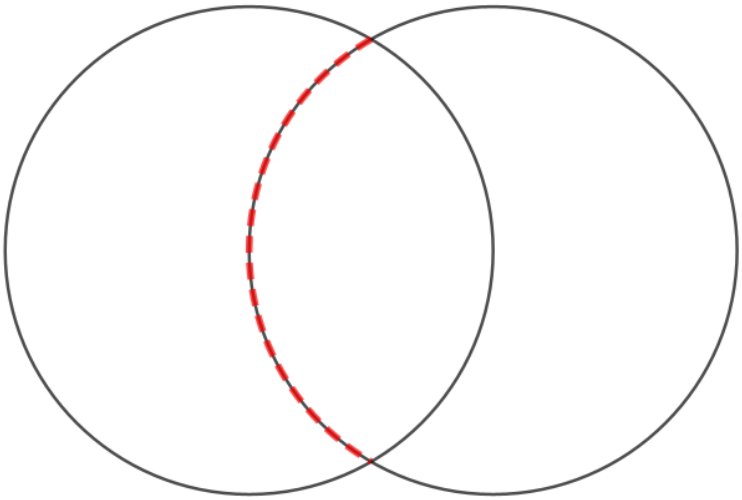
It has been made clear that $z_1 = e^{i\alpha_1}$ will lie on the unit circle centered at the origin. Put another way, the range of z_1 is the unit circle, $|z_1| = 1$ for all possible values of z_1. This is why I have not yet considered the question of what the value of $P(|z_1| < 1)$ is, because $P(|z_1| < 1) = 0$ (if we reframed the question to being, *what is the probability that z_n lies **on** or within the unit circle?*, then $P(|z_1| \leq 1) = 1$, but I specifically thought of question in terms of the open set of $\{|z| < 1, z \in \mathbb{C}\}$ - this is quite arbitrary and cannot be justified, but my intention is not to do so; rather, it is to make explicit that $P(|z_1| < 1) = 0$). However, the only special characteristic of z_1 is that it is the first number in any of the potential series we could construct for z_n. It is the significance of the Euler and polar forms of complex numbers that the numbers lie on a circle when the modulus is constant. Every $z_x = e^{i\alpha_x}$ will

lie on a circle of radius 1, only with different centers.

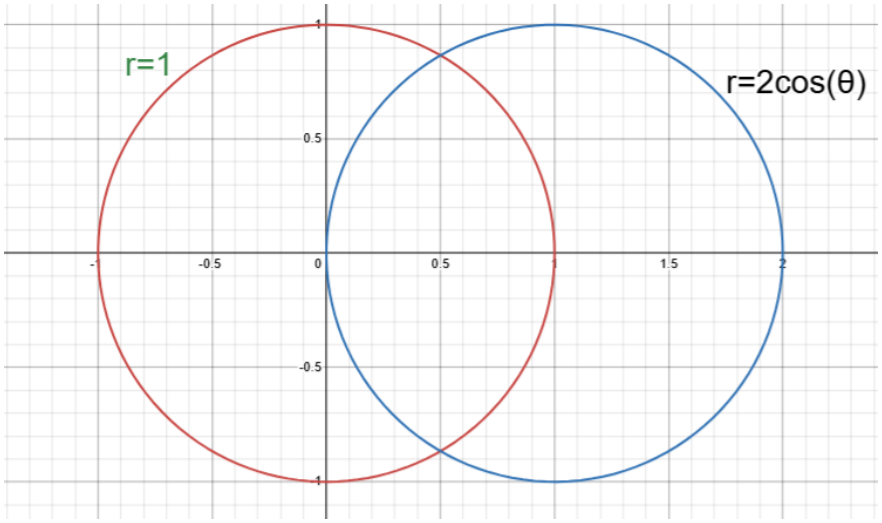
This becomes obvious in an adjusted Desmos model of z_2, this one displaying the range of values for the first and second constituent complex numbers of z_2:



When I made this model to explore what z_2 behaves like geometrically, I realized that, for this z_1, z_2 landing in the unit circle depends completely on the second number landing on the arc segment of the circle describing its range that is within the first circle, regardless of where the first number lands, because z_1 will always lie on the unit circle centered at the origin. This arc segment is highlighted in the diagram below. Thus, the $P(|z_2| < 1)$ for this z_1 is the ratio of the length of that arc segment to the circumference of the second circle. Let this ratio be called S. Therefore, $P(|z_2| < 1) = S$ for this z_1. And because the ratio S does not depend on the value of z_1, $P(|z_2| < 1) = S$ for all z_1.



I thus begin finding the length of the sector. Consider the polar functions $r = 1$ and $r = 2\cos(\theta)$, which are functional representations of the diagram above, graphed below in the polar plane:



We see from the graph that the θ value of intersections of these functions will lie between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, but let us set them equal to each other and solve for the intersection:

$$1 = 2\cos(\theta)$$

$$\frac{1}{2} = \cos(\theta)$$

$$\theta = \cos^{-1}(\frac{1}{2})$$

$$\theta = -\frac{\pi}{3}, \frac{\pi}{3}$$

Therefore, $r = 1$ and $r = 2\cos(\theta)$ intersect when $\theta = -\frac{\pi}{3}$ and $\theta = \frac{\pi}{3}$. For $-\frac{\pi}{3} < \theta < \frac{\pi}{3}$, $r = 2\cos(\theta)$ will actually fall outside the unit circle such that $r > 1$ because a positive change in θ will cause a counter-clockwise movement along the circle, but this tells us that there is a change in θ of magnitude $\frac{\pi}{3} - (-\frac{\pi}{3}) = \frac{2\pi}{3}$ that will cover on $r = 2cos(\theta)$ the arc segment that we seek. S is then this change in θ, $\Delta\theta = \frac{2\pi}{3}$ divided by the circumference of a unit circle. The circumference of a circle is given as $C = 2\pi r$, so

$$S = \frac{\left(\frac{2\pi}{3}\right)}{2\pi} = \frac{1}{3},$$

and since $S = P(|z_2| < 1)$, $P(|z_2| < 1) = \frac{1}{3}$, as suggested by the earlier hypothesis.

When I first arrived at this point, the result felt quite serendipitous, like I just got lucky with how z_2 relates to circles. Once that connection is made and the intuition is built via the simulations, the $\frac{1}{3}$ feels obvious, but the fact that it would work out in this way is still somewhat surprising to me. Starting with just the complex numbers, or thinking of the question as something like finding the value of (the equivalent, but perhaps initially unhelpful, expression) $P(\sqrt{(\cos(\alpha_1) + \cos(\alpha_2))^2 + (\sin(\alpha_1 + \sin(\alpha_2)))^2} < 1)$, I never would have suspected that $\frac{1}{3}$ is the answer. It is by the models and geometric connections that it became clearer, and then, at least retrospectively, obvious. It is verified that $P(|z_2| < 1) = \frac{1}{3}$, and in the circle model, it makes a lot of sense. Consider how the consideration of the ratio of the segment and the circumference led to a correct hypothesis whereas the earlier hypothesis, that proposed that the probability is $\frac{1}{4}$ because every point within a circle of radius 2, centered at the origin, is equally likely as a value of z_2, was incorrect. I believe it was in developing better models of z_2 that the hypothesis developed.

Redirecting our attention to the goal of the paper, this section has answered the first question posed in the introduction, and one case of the ultimate question. Now we move on to thinking about higher values of n for z_n.

2.3 $P(|z_n| < 1)$ for Higher Values of n

2.3.1 Simulation

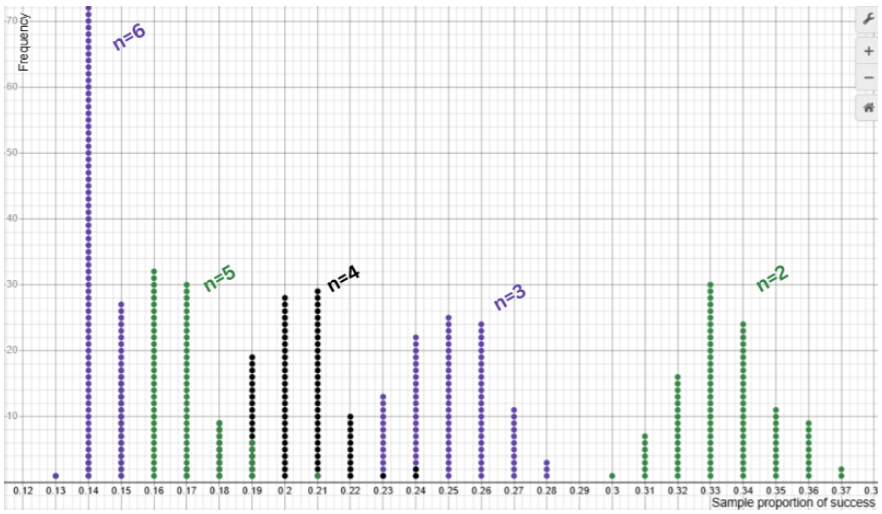
To follow with the original goal of attempting to use and generalize the process of finding the probability of success when $n = 2$, let us begin once more with a simulation. I adjusted `IA1.py` to calculate the proportion of successes for any number of constituent complex numbers; that is, this program, named `IA3.py` and found in §A.3, will take n as an input (the variable is `b` in the program, because `n` was already being used for indexing), and estimate the proportion of success for that specific z_n.

Consider this example output for $n = 3$ (here, the program is taking 100 samples):

```
PS C:\Users\samre\OneDrive\Desktop\project> python .\IA3.py
How many complex numbers?3
[0.2467, 0.2521, 0.2486, 0.2498, 0.2527, 0.256, 0.2527, 0.2507, 0.2532, 0.241, 0.2514,
0.2522, 0.2496, 0.2489, 0.2398, 0.2457, 0.2461, 0.2505, 0.2536, 0.2484, 0.2528, 0.2475,
 0.2454, 0.2511, 0.256, 0.2526, 0.2432, 0.2479, 0.2449, 0.2512, 0.2516, 0.2507, 0.2592,
 0.2381, 0.2473, 0.2564, 0.2438, 0.2565, 0.2533, 0.2515, 0.2475, 0.2501, 0.2543, 0.2462
, 0.2503, 0.2518, 0.2451, 0.2403, 0.2577, 0.2433, 0.2504, 0.2465, 0.2558, 0.2502, 0.251
2, 0.2573, 0.2491, 0.2481, 0.2501, 0.2567, 0.249, 0.2511, 0.2501, 0.2476, 0.2525, 0.256
5, 0.2479, 0.2518, 0.2462, 0.2535, 0.2486, 0.2454, 0.2488, 0.2511, 0.2482, 0.2516, 0.24
29, 0.2446, 0.2469, 0.2585, 0.2441, 0.255, 0.2507, 0.251, 0.2558, 0.2422, 0.2487, 0.257
2, 0.2565, 0.2527, 0.2532, 0.2489, 0.2546, 0.2503, 0.2542, 0.2468, 0.2461, 0.2429, 0.24
38, 0.2543]
```

This too can be plotted in order to make a more reliable estimate, but before we construct the dot plot, notice how the proportion of successes for $n = 3$ seems to be $\frac{1}{4}$.

Since we are not necessarily focused on the value of $P(|z_n| < 1)$ for one particular value of n, I used `IA3.py` to give sample proportions for multiple values of n and plotted them all at once using the same process shown above. For $n = 2$ through $n = 6$, I ran the simulation and plotted the data:



The means and medians of these data sets, the 100 sample proportions for each value of n, as calculated by Desmos are shown in the table below:

n	2	3	4	5	6
mean	0.33422	0.24967	0.20106	0.16437	0.142694
median	0.333	0.251	0.201	0.1635	0.14275

Now, there is a pattern in these numbers. We have already established that $P(|z_2| < 1) = \frac{1}{3}$ and hypothesized that $P(|z_3| < 1) = \frac{1}{4}$. The table above seems to suggest, by estimation but lacking in proof, that $P(|z_4| < 1) = \frac{1}{5}$, $P(|z_5| < 1) = \frac{1}{6}$, and so on. So I introduce a new hypothesis:

$$P(|z_n| < 1) = \frac{1}{n+1}$$

when $n > 1$, $n \in \mathbb{Z}^+$ because if $n = 1$, the above formula would give $P(|z_1| < 1) = \frac{1}{2}$. But it has been established that $P(|z_1| < 1) = 0$ and $P(|z_1| \leq 1) = 1$. There is no way for $P(|z_1| < 1)$ to equal $\frac{1}{2}$. And any value of n less than 1 is meaningless given our definition of z_n.

However, I do not know how to prove (or disprove) this hypothesis, and have not been able to make any advances on it. So we move on to the conclusion of the paper.

3 Conclusion

3.1 Reflections

The clearest purpose of this conclusion is to evaluate to what extent I achieved my goal. I see in this two questions that need answering: (1) to what extent did I complete the task and (2) to what extent were my methods and investigations the best?

To answer the first question, recall the aim of the investigation, which was to answer the question:

What is the probability that z_n falls within the unit circle $|z| = 1$ for a given n?

Now, I did prove that $P(|z_2| < 1) = \frac{1}{3}$, but that is only a special case of the question. And then, I gave a hypothesis that, if proven, would achieve the aim in totality, but I have not proved it. Thus, I can only leave a conjecture as my answer, and nothing more substantial. The investigation is incomplete.

Below, I'll consider a few more ideas that could be followed to continue the investigation, but are not part of this project. Before I do, however, I will first evaluate the efficiency of the investigation. Many of the decisions I made in how I went about this were quite arbitrary, but I think that is due to the fact that the whole question was also arbitrary. For example, why did I focus so much on circles? Well, I thought of the question, at least in part, in relation to circles. Also, because I came up with the question that was to be answered, there was, from the start, no guarantee that I would be able to completely answer it. These are, perhaps, the dangers of exploring such an abstract and self-developed (in the sense that this was "self-developed") topic. Perhaps it would have been better, and made for a more logical and successful investigation to have worked harder to develop a better question upon which the IA would be based. Additionally, I relied heavily on modeling and simulation, which are important parts of mathematical inquiry, but did not perform as much analytical, computational, and even just symbolic math as I would have liked. It could be that this was just because of how I approached it; that is, the paradigm in which my thought was stuck

necessitated simulation. On the topic of such a paradigm, I could have, I think, benefited from looking at the problem through more and different perspectives. For example, the case where $n = 2$ becomes quite manageable when thinking about circles, but circles are not conducive for the $n > 2$ cases.

3.2 Further Investigation

Then this leads to asking how, if I had the time, or if this IA was expanded to include more, I would consider the $n > 2$ cases. My first thought is to find probability distributions that model the values of z_n for each n. Maybe this could be done by having some sort of complex-valued random variable or probability density curves (or surfaces) in higher dimensions. Or, taking the conjecture that $P(|z_n| < 1) = \frac{1}{n+1}$, I could research equations, formulae, and functions that also equal $\frac{1}{n+1}$, or the equivalent, and then work backwards and find some connection between that found equation and the complex numbers here. But for now, and for this project, this is where the investigation ends.

References

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Appendix

A Code

A.1 IA1

```
#makes 50 random samples of size 1000 of the 2
complex number case and outputs proportions of
successes
#and a list of these proportions


#good data for dotplot


import random
import math


n=0
m=0
```

```
a=0
list=[]


while m<50:
 while n<1000:
  alpha1=random.random()*2*math.pi
  alpha2=random.random()*2*math.pi
  v=math.sqrt((math.cos(alpha1)+math.cos(alpha2))**2 +
  (math.sin(alpha1)+math.sin(alpha2))**2)
  if v<1:
   a=a+1
  n=n+1
 print(a/n)
 list.append(a/n)
 m=m+1
 n=0
 a=0
print(list)
```

A.2 IA3

```
#simulation for higher number complex number cases-->
#basically an upgraded version of IA1.py


import random
import math
import sys
```

```
m=0 #number of samples taken of size n
n=0 #number of trials in a sample (sample size)
a=0 #count of successes in one sample = no. of times final
#complex number lies inside unit circle (|z_net|<1)

b=int(input("How many complex numbers?")) #allows user to
#select amount of complex numbers to add

successes=[] #list of proportions of success for all
#samples, has m elements, convenient for plotting

while(m<100): #run m samples
 while(n<10000): #run n trials
  real_net=0 #real part of net complex number
  imaginary_net=0 #imaginary part of net complex number
  i=0 #index
  while(i<b): #iterate process of adding random complex numbers b times
   alpha_i=random.random()*2*math.pi #random angle
   real=math.cos(alpha_i) #real part of specific,
   #indexed complex number (not net)
   imaginary=math.sin(alpha_i) #imaginary part
   #of specific, indexed complex number (not net)
   real_net=real_net+real
   imaginary_net=imaginary_net+imaginary
   i=i+1
  #find modulus of net complex number
  z_net=math.sqrt((real_net**2)+(imaginary_net**2))
```

```
  #if z_net<1 is used here, b=1 will not work because sometimes,
  #what should be outputted as 1 is 0.9999999999999999, so this
  #condition must allow for error If success (|z_net|<1),
  if z_net<0.999999999999999:
   a=a+1 #add one to count (a)
  n=n+1

 print(a/n)
 successes.append(a/n)
 m=m+1
 n=0
 a=0
print(successes) #prints all sample proportions in the m samples
```